MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC ASSESSMENT TEST – 10 (LINEAR EQUATIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 3: Pair of Linear Equations in Two Variables** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 14(iv).
2. **Sec A:** 4 MCQs (1M). **Sec B:** 2 AR (1M). **Sec C:** 3 VSAQ (2M). **Sec D:** 3 SA (3M). **Sec E:** 1 LA (5M). **Sec F:** 1 Case Study (4M).

SECTION A: Multiple Choice Questions (4 × 1 = 4 Marks)

Q1. If the system of equations $3x + y = 1$ and $(2k-1)x + (k-1)y = 2k+1$ is inconsistent, then k equals: [1]
(A) -1 (B) 0 (C) 1 (D) 2

Q2. The coordinates of the point where the line $2x - 3y = 12$ intersects the x-axis are: [1]
(A) $(0, -4)$ (B) $(6, 0)$ (C) $(-6, 0)$ (D) $(0, 4)$

Q3. If $ax + 2y = 7$ and $3x + by = 18$ represent parallel lines, where a and b are integers, then ab equals: [1]
(A) 5 (B) 6 (C) 7 (D) 9

Q4. A system of two linear equations in two variables is said to be dependent if it has: [1]
(A) no solution (B) exactly one solution (C) infinitely many solutions (D) exactly two solutions

SECTION B: Assertion-Reason Questions (2 × 1 = 2 Marks)

Directions: (A) Both (A) and (R) are true, and (R) is correct explanation. (B) Both true, but (R) is not correct explanation. (C) (A) is true, (R) is false. (D) (A) is false, (R) is true.

Q5. **Assertion (A):** The solution of the pair of linear equations $x + y = 3$ and $2x + 5y = 12$ is $x = 1, y = 2$.
Reason (R): A pair of values (x, y) that satisfies one equation of a system will automatically satisfy any other equation. [1]

Q6. **Assertion (A):** If the perimeter of a rectangle is 30 cm and its length is 3 cm more than twice its breadth, then its breadth is 4 cm.
Reason (R): The algebraic formulation is $2(l + b) = 30$ and $l = 2b + 3$. [1]

SECTION C: Very Short Answer Questions (3 × 2 = 6 Marks)

Q7. Solve $2x + 3y = 11$ and $2x - 4y = -24$, and hence find the value of m for which $y = mx + 3$. [2]

Q8. The larger of two supplementary angles exceeds the smaller by $18°$. Find the angles. [2]

Q9. A fraction becomes $1/3$ when 1 is subtracted from numerator and $1/4$ when 8 is added to denominator. Find the fraction. [2]

SECTION D: Short Answer Questions (3 × 3 = 9 Marks)

Q10. Solve for x and y: $152x - 378y = -74$ and $-378x + 152y = -604$. [3]

Q11. Yash scored 40 marks in a test, getting 3 marks for each right answer and losing 1 mark for each wrong answer. Had 4 marks been awarded for each correct answer and 2 marks been deducted for each incorrect answer, Yash would have scored 50 marks. How many questions were there in the test? [3]

Q12. A man sold a chair and a table together for Rs. 1520 making a profit of 25% on the chair and 10% on the table. By selling them together for Rs. 1535 he would have made a profit of 10% on the chair and 25% on the table. Find the cost price of each. [3]

SECTION E: Long Answer / HOTS Question (1 × 5 = 5 Marks)

Q13. A part of monthly hostel charges in a college is fixed and the remaining depends on the number of days one has taken food in the mess. When student A takes food for 20 days, he has to pay Rs. 1000 as hostel charges whereas student B, who takes food for 26 days, pays Rs. 1180. Find the fixed charges and the cost of food per day. If student C takes food for 28 days, find the total charges he has to pay. [5]

SECTION F: Applied Case Study Based Question (1 × 4 = 4 Marks)

Q14. Case Study: Competitive Examination Evaluation & Negative Marking

In an Olympiad scholarship examination of 60 questions, candidate marks are calculated under a differential scoring scheme. A student receives 4 marks for every correct answer, while 1 mark is deducted for every incorrect response. Kabir attempted all 60 questions and scored a net total of 140 marks. Let x denote the number of correct answers and y denote the number of incorrect answers.

Based on the above examination scoring criteria, answer the following questions:

(i) Write the system of linear equations in x and y representing the total questions attempted and Kabir's score. [1]

(ii) Determine how many questions Kabir answered correctly (x). [1]

(iii) Determine how many questions Kabir answered incorrectly (y). [1]

(iv) If negative marking is increased to 2 marks per wrong answer, what would Kabir's score be for the same correct answers? [1]

OR

Find the minimum number of correct answers needed to score at least 180 marks with 1 mark penalty. [1]

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